

HEPRO AI+

Activity - 2

Objective

The objective of this activity is to convert raw student data into meaningful, interpretable performance scores that mentors can easily understand and act upon. The scoring system uses deterministic, rule-based logic to ensure transparency, consistency, and explainability in student evaluation.

Input

students.csv

A structured dataset containing student academic, wellness, productivity, engagement, and career-related attributes.

Scoring Dimensions Implemented

The following core scores are computed for each student. All scores are normalized to a **0–100 scale**.

1. Academic Performance Score (APS)

Purpose:

Measures academic consistency and performance.

Features Used:

- GPA
- Attendance percentage
- Assignment completion percentage

Formula:

$$\text{APS} = (\text{GPA} \times 10 \times 0.5) + (\text{Attendance} \times 0.3) + (\text{Assignments Completion} \times 0.2)$$

HEPRO AI+

2. Wellness & Wellbeing Score (WWS)

Purpose:

Evaluates mental and physical wellbeing.

Features Used:

- Stress level (inverse relationship)
- Sleep hours
- Mental wellbeing score

Formula:

$$\text{WWS} = ((10 - \text{Stress Level}) \times 10 \times 0.4) + (\text{Sleep Hours} \times 10 \times 0.3) + (\text{Mental Wellbeing} \times 10 \times 0.3)$$

3. Productivity & Time Management Score (PTMS)

Purpose:

Measures efficiency, focus, and time management.

Features Used:

- Productivity score
- Distraction level (inverse relationship)

Formula:

$$\text{PTMS} = (\text{Productivity Score} \times 10 \times 0.5) + ((10 - \text{Distractions}) \times 10 \times 0.5)$$

HEPRO AI+

4. Career Readiness Score (CRS)

Purpose:

Assesses preparedness for career planning and skill development.

Features Used:

- Career clarity
- Skill readiness

Formula:

$$\text{CRS} = (\text{Career Clarity} \times 10 \times 0.5) + (\text{Skill Readiness} \times 10 \times 0.5)$$

Student Readiness Index (SRI)

Purpose:

Acts as a single, comprehensive indicator of overall student readiness.

Formula:

$$\begin{aligned} \text{SRI} = & 0.30 \times \text{APS} \\ & + 0.25 \times \text{WWS} \\ & + 0.20 \times \text{PTMS} \\ & + 0.25 \times \text{CRS} \end{aligned}$$

The SRI score is used as the **primary trigger for mentoring interventions**.

HEPRO AI+

Risk Classification Logic

Students are categorized based on their SRI score to enable quick mentor decision-making.

SRI Range	Category	Interpretation
≥ 80	Green	High performing and stable
65 – 79	Blue	Good performance, minor support needed
50 – 64	Yellow	Requires attention and guidance
< 50	Red	High risk, urgent mentoring required

Validation

Sample student records were evaluated to verify:

- Correct score computation
- Logical category assignment
- Realistic differentiation between high-performing and at-risk students

The results align with expected real-world student behavior.